

Solar Tracking Project

Using ARDUINO UNO

TEAM MEMEBERS:

- PRIYADHARSHINI G
- SHALINI S V
- ANADHALAKSHMI G

Objectives

To design a system that automatically tracks sunlight

To measure light intensity using two LDR sensors

To process sensor data using
ARDUINO UNO

To rotate a solar panel using a servo motor toward maximum light

To improve the efficiency of solar energy collection

Block Diagram



Light detector
(X direction)

Comparator

Light detector
(Y direction)

X-direction
motor

Motor Driving
Circuit

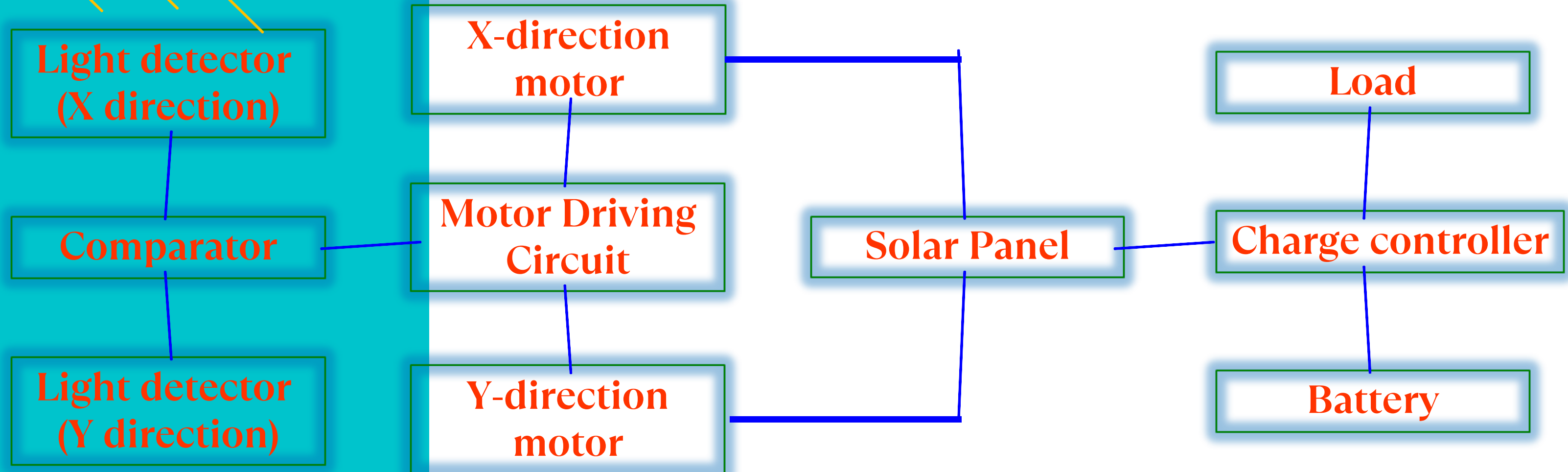
Y-direction
motor

Solar Panel

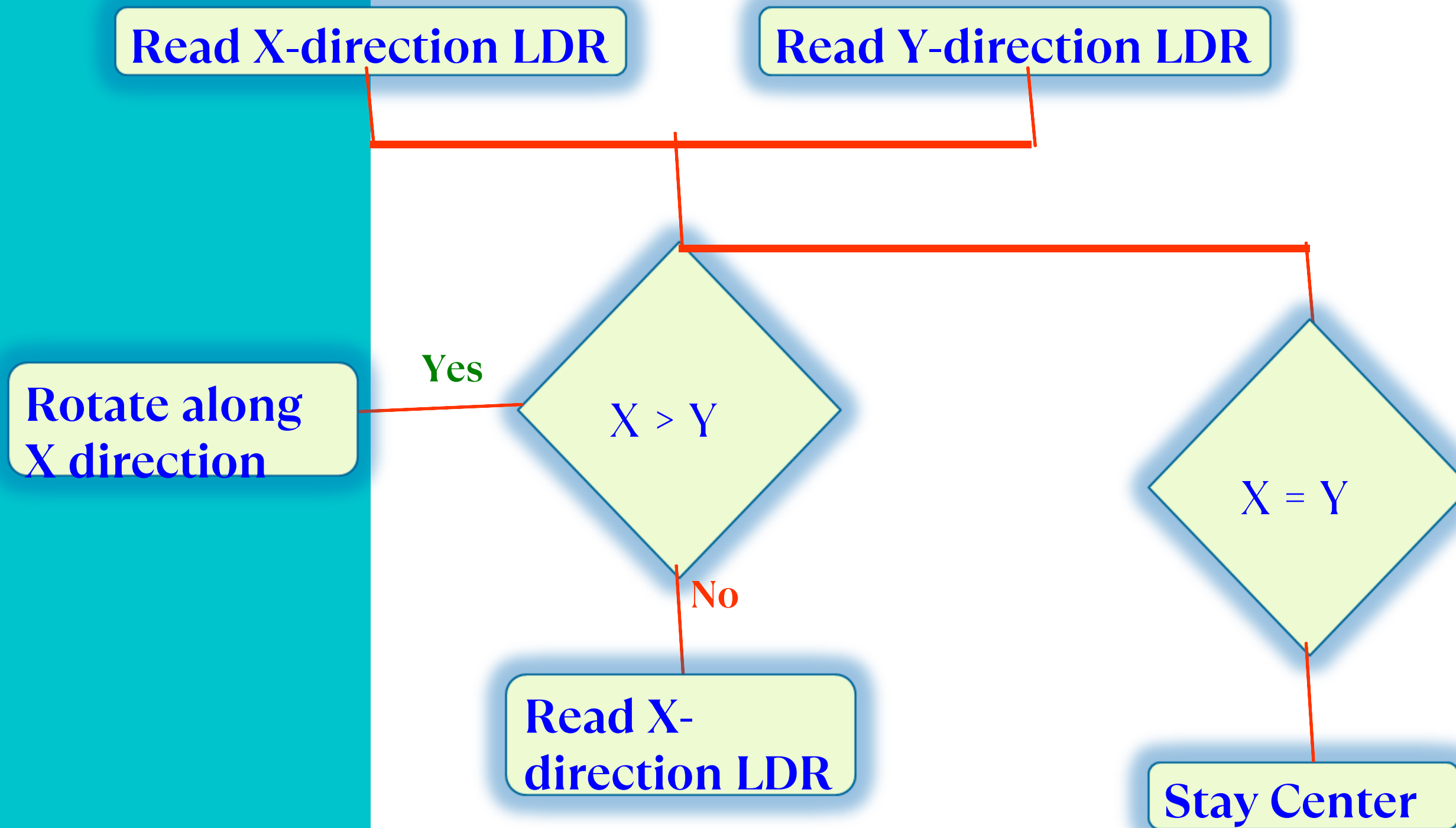
Load

Charge controller

Battery



Flow Chart



Hardware Used

Hardware Used

1

Arduino Uno

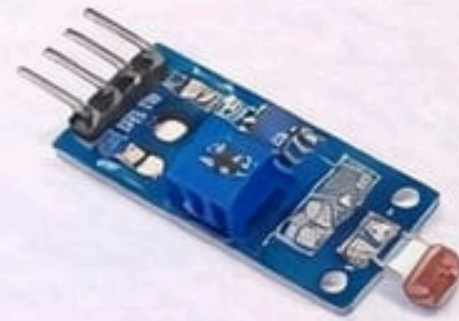
- Microcontroller based on ATmega328P
- Operating voltage: 5V
- 14 Digital I/O pins (6 PWM)
- 6 Analog input pins
- 10-bit ADC



2

LDR Sensors

- Detects light intensity
- Resistance decreases with light
- Used in voltage divider



3

SG90 Servo Motor

- Operating voltage: 4.8V – 6V
- Rotation: 0°–180°



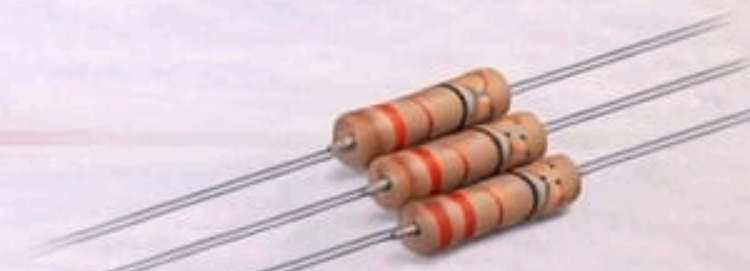
4

Solar Panel

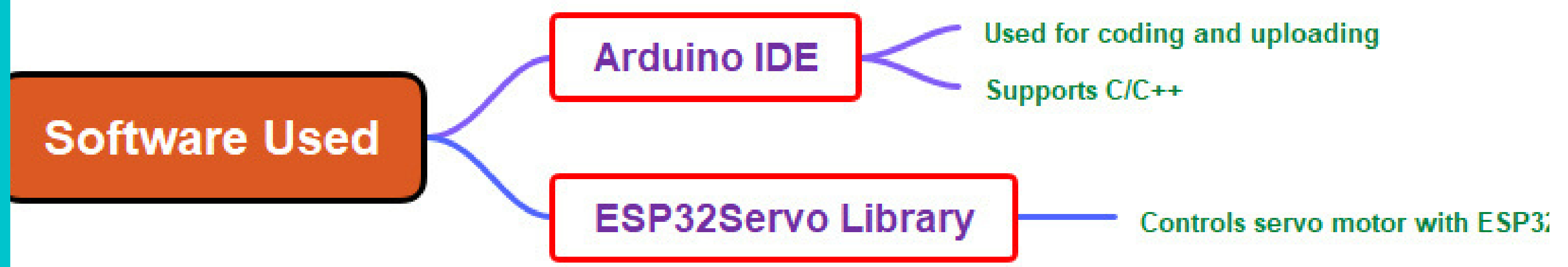
- Converts sunlight into electricity

5

Resistors (10k Ω)



Software Used



Budget

Component	Quantity	Amount
ARDUINO UNO	1	1000.00
LDR Sensors	2	40.00
Servo Motor	1	150.00
Solar Panel	1	200.00
Resistors/Wires	-	50.00
	Total	1440.00

Outcome / Results

- Panel automatically tracks sunlight
- Compares light intensity accurately
- Servo adjusts position effectively
- Low-cost implementation
- Improves energy efficiency

Team Contribution

- All Team Members Contributed Equally
- All members actively participated in idea development and planning
- Worked together on circuit design and hardware connections

Conclusion

- Successfully implemented automatic solar tracking system
- Improved the solar panel efficiency
- Cost-effective and simple design
- Can be extended to dual-axis tracking
- Possible to make commercially available in bulk size

THANK YOU